

# ■ COMPUTER NETWORKS — Complete Learning Mind Map

A connected study roadmap for assessments, interviews, and practical software engineering

## Priority Legend

■ HIGH PRIORITY — Must understand deeply   ■ MEDIUM PRIORITY — Important for assessments/interviews   ■ LOWER PRIORITY — Learn basics first, deepen later   ■ CONNECTION POINT — Connects multiple topics

## 1. NETWORKING FUNDAMENTALS ■

- Computer Network → Devices communicating and sharing data/resources
- Internet → Network of Networks ■
- Why networking? → Communication, resource sharing, remote access, distributed applications
- Types → PAN, LAN ■, MAN, WAN
- Topologies → Star, Bus, Ring, Mesh, Hybrid
- Protocol → Rules that allow devices to communicate

## 2. NETWORK MODELS & LAYERS ■■■

- Why layers exist → Divide complex communication into responsibilities; each layer solves a specific problem; layers work together
- OSI Model → Application, Presentation, Session, Transport ■, Network ■, Data Link ■, Physical ■
- TCP/IP Model → Application → Transport → Internet/Network → Network Access

## 3. DATA FLOW & ENCAPSULATION ■■■

- Sender flow: Application DATA → Transport SEGMENT → Network PACKET → Data Link FRAME → Physical BITS
- Encapsulation → Adding headers/information while moving downward
- Decapsulation → Removing headers/information at the receiver

## 4. TRANSPORT LAYER ■■■■

- Main purpose → Process-to-Process communication
- Ports ■ → Identify application/service; Source Port vs Destination Port
- Common ports → HTTP 80, HTTPS 443, SSH 22, DNS 53, SMTP 25
- TCP ■ → Connection-oriented, reliable, ordered delivery
- TCP topics → 3-way handshake, termination, sequence numbers, acknowledgements, retransmission, flow control, congestion control
- UDP ■ → Connectionless, lower overhead, no guaranteed delivery
- UDP use cases → Streaming, gaming, DNS, real-time communication

## 5. NETWORK LAYER ■■■■

- Main purpose → Communication across different networks
- IP Addressing → IPv4 ■, IPv6 ■, Public IP ■, Private IP ■

- IP Packet → Source IP + Destination IP
- Routing → Routing table, next hop, default route
- Router → Connects different networks, examines destination IP, forwards packets

## 6. LOCAL NETWORKS (LAN) ■■■■

- LAN → Devices communicating within a local area
- Same network → Device can communicate directly through switch/Wi-Fi
- Different network → Device sends packet to Default Gateway
- Default Gateway ■ → Router used to reach outside networks
- KEY DECISION: Is destination inside my local network? YES → Send directly | NO → Send to gateway

## 7. SUBNETTING & SUBNET MASK ■■■

- Why subnetting? → Divide networks into smaller networks
- Subnet mask → Separates Network Part and Host Part
- CIDR notation → /8, /16, /24
- Key skill → Determine whether two IP addresses belong to the same network

## 8. DATA LINK LAYER ■■■

- MAC Address ■ → Local interface / hop-to-hop communication
- Frame → MAC header + trailer
- Switch ■ → Uses MAC address table to forward frames
- ARP ■ → 'I know an IP address; what is the corresponding MAC address?'
- Ethernet / Wi-Fi ■

## 9. PHYSICAL LAYER ■

- Bits transmitted as electrical, radio, or optical signals
- Media → Ethernet cable, fiber optics, wireless
- Basic understanding is enough initially

## 10. ADDRESSING — THE BIG CONNECTION ■■■■

- PORT NUMBER → Transport Layer → Which application/process?
- IP ADDRESS → Network Layer → Which destination host?
- MAC ADDRESS → Data Link Layer → Which next-hop interface?
- Core connection: Port → Application | IP → Host | MAC → Next-hop local interface

## 11. APPLICATION LAYER ■■■■

- HTTP/HTTPS → Client-server model, request/response, methods (GET/POST/PUT/PATCH/DELETE), status codes
- DNS ■ → Domain Name → IP Address
- DHCP ■ → Provides IP address, subnet mask, default gateway, DNS server
- Other protocols → FTP, SMTP, POP3/IMAP, WebSocket

## 12. DNS — DEEP UNDERSTANDING ■■■■

- Why DNS? → Humans remember names better than IP addresses
- Basic flow → Domain name → DNS resolver → lookup → IP address
- Hierarchy → Root → TLD → Authoritative server
- DNS caching

## 13. NAT ■■■■

- Why NAT? → Private IP addresses are not globally routable
- Home device private IP → Router NAT → Public Internet IP
- NAT translation table helps responses return to the correct internal device

## 14. NETWORK DEVICES ■■■

- Hub ■ → Physical Layer, broadly repeats signals
- Switch ■ → Data Link Layer, MAC-based forwarding
- Router ■ → Network Layer, IP-based forwarding
- Modem ■ → Connects to ISP infrastructure
- Access Point ■ → Provides wireless access
- Firewall ■ → Filters network traffic
- Proxy ■ → Intermediary between client and server

## 15. NETWORK SECURITY BASICS ■■■

- HTTP vs HTTPS ■
- TLS/SSL ■ → Encryption during communication
- Firewall ■
- VPN ■
- Authentication vs Authorization ■
- Basic attacks → Man-in-the-Middle, DDoS, Spoofing

## 16. RELIABILITY & PERFORMANCE ■

- Latency → Time taken for data to travel
- Bandwidth → Maximum data capacity
- Throughput → Actual data transferred
- Packet Loss
- Jitter → Variation in packet arrival time

## ■ Recommended Study Order

### Phase 1 — Foundation ■

Networking Basics → Why Layers? → OSI vs TCP/IP → Encapsulation → Port vs IP vs MAC

### Phase 2 — Local Networking ■

LAN → IP Address Basics → Subnet Mask → Same vs Different Network → Default Gateway → MAC → ARP → Switch → Router

### Phase 3 — Transport Layer ■■

Ports → TCP vs UDP → TCP Connection → 3-Way Handshake → Reliability → Sequence Numbers → ACK → Flow Control → Congestion Control

### Phase 4 — Network Layer ■■

IPv4 → Public vs Private IP → Subnetting → Routing → Routing Table → Default Gateway → NAT

### Phase 5 — Application Protocols ■

DNS → HTTP → HTTPS → DHCP → Email Protocols → WebSocket basics

## ■ The Most Important Conceptual Connections

### 1. Addressing

Port → Which application? | IP → Which host? | MAC → Which next-hop interface?

### 2. Local vs Internet

Destination IP → Same Network? YES → ARP for destination MAC; NO → ARP for default gateway MAC

### 3. Sending Data

Application Data → TCP/UDP → IP → MAC → Bits

### 4. Internet Journey

Device → Switch/Wi-Fi → Default Gateway → NAT → ISP → Multiple Routers → Destination Network → Server

## ■ COMPLETE REAL-WORLD FLOW: Opening <https://example.com>

- 1. Application: Browser wants to reach example.com
- 2. DNS: Domain name is resolved to an IP address
- 3. Transport: Source and destination ports identify the communication endpoints
- 4. Network: Source and destination IP addresses are placed in the IP packet
- 5. Local decision: Same network? Send directly. Different network? Send to default gateway
- 6. ARP: Find the MAC address of the required local next hop
- 7. Data Link: Create a frame with source and destination MAC addresses
- 8. Physical: Convert bits into signals
- 9. Routers: Remove incoming frame → inspect destination IP → choose next hop → create new frame → forward
- 10. Destination server: Decapsulation delivers the data upward to the application

## ■ Final Study Advice

Do not memorize Computer Networks as isolated definitions. For every topic, ask: (1) What problem does it solve? (2) Which layer does it belong to? (3) What other concepts does it connect with? Build understanding through real-world journeys such

as opening a website, sending a WhatsApp message, or making an API request.